

SHARIAGUARD: Scholar-Led, AI-Screened, and Blockchain-Enforced Shariah Compliance for Every *Akad* in Islamic Fintech

"Guarding trust, one akad at a time."

Abstract

Islamic fintech is growing quickly, and with it the need for Shariah supervision that is both scalable and transparent. Today, however, checking whether *akad* (contracts) comply with Shariah is still largely done by hand, which leads to delays, uneven judgements, and a risk of non-compliance. The essay introduces SHARIAGUARD, a governance framework in which artificial intelligence and blockchain smart contracts work together to check and enforce *akad* compliance automatically. Its three layers are an NLP screening module built around DSN-MUI fatwas, smart contracts that hold approved conditions in place, and an audit dashboard that supervisory boards and regulators can consult at any time. Throughout, the framework is shaped by *Maqasid Shariah* and by three ethical AI commitments, namely fairness, transparency, and accountability, while the final ruling always rests with human scholars. In line with the vision of Society 5.0, the framework contributes to Sustainable Development Goals (SDGs) 16, 9, and 8. The essay closes with an implementation roadmap and policy recommendations to strengthen trust and financial inclusion in the Islamic economy.

Keywords: Islamic fintech, digital Shariah governance, ethical AI, smart contract, akad compliance, SDGs, Society 5.0

1. Introduction

Islamic fintech has emerged as a channel that promises to extend Shariah-compliant financing, payments, and investment to people who are underserved by conventional institutions. Digital platforms can originate, price, and settle a transaction within minutes, and industry reports describe the Islamic economy as a sector that keeps expanding (Dinar Standard, 2023). That speed, however, brings a governance problem. A product counts as Islamic only if each of its contracts (*akad*) meets Shariah requirements, so a platform that cannot show this has little left to offer the customers who came to it for that reason.

Shariah supervision has traditionally been the job of a supervisory board, known in Indonesia as the *Dewan Pengawas Syariah* (DPS). The board typically reviews product structures, contract templates, and a sample of transactions. That works well enough for an institution with only a handful of products. A digital platform is a different story. Large

volumes of standardized contracts are issued every day, each with terms adjusted to the individual customer, and a review done at intervals cannot keep up. Records also end up spread across several systems, so anyone outside the board has little chance of tracing how a given decision was made.

Digital technology created this pressure, but it may also help relieve it. Artificial intelligence (AI) can read and classify contract text at scale, and blockchain can fix agreed terms in a tamper-resistant record (Nakamoto, 2008). Such tools are central to the Industry 4.0 and Society 5.0 agendas, which seek to combine digital technology with human-centered social outcomes (Cabinet Office, Government of Japan, n.d.). However, adopting these tools without Shariah safeguards would risk replacing scholarly judgement with opaque algorithms, which contradicts the purpose of Islamic governance. This essay therefore asks one central question: how can AI and blockchain be integrated into a digital Shariah governance framework that automates *akad* compliance while preserving scholarly authority and ethical safeguards?

To answer it, the essay proposes SHARIAGUARD, with the tagline "Guarding trust, one *akad* at a time." It is a three-layer framework that combines an AI-based screening module, smart contract enforcement, and an audit dashboard, with the DPS positioned as the final authority. SHARIAGUARD is intended to contribute to the United Nations Sustainable Development Goals (SDGs) (United Nations, 2015). Its main link is SDG 16, which concerns strong and trustworthy institutions, and it also touches SDG 9 on innovation and SDG 8 on inclusive economic growth. The rest of the essay is organized as follows. Section 2 examines where current compliance practice falls short, Section 3 lays out the conceptual basis, and Section 4 describes the design and the wider solution package. Ethical AI is discussed in Section 5, implementation in Section 6, and the link to the SDGs in Section 7, before Section 8 concludes.

2. Problem Analysis

Three gaps in current practice motivate the proposed framework. Table 1 summarizes how each gap is addressed by SHARIAGUARD.

The first gap is manual and inconsistent checking, because each type of *akad* carries its own validity conditions. In Indonesia, the National Sharia Council of the Indonesian Ulema Council (DSN-MUI) has issued fatwas for specific contracts such as *murabahah* (DSN-MUI, 2000a) and *mudharabah* (DSN-MUI, 2000b), as well as for technology-based financing services (DSN-MUI, 2018). A *murabahah* contract, for instance, requires clear disclosure of

the cost price and the profit margin, whereas a *mudharabah* contract needs a profit-sharing ratio agreed in advance. Turning such conditions into checks on every single contract by hand is slow, and two reviewers may reach different results. As the number of contracts grows, manual sampling covers an ever smaller share of the total.

The second gap is hidden non-compliance. Digital contracts are usually standardized and accepted with a single click, so a problematic clause is easy to miss. Margins may be stated so loosely that they drift toward *riba*. Terms may be vague enough to create *gharar* (excessive uncertainty). Late-payment charges may be booked as platform income, even though Shariah principles generally expect them to go to charity. Terms can also be changed after approval, especially when the records sit in a system controlled by one party. Because these problems are buried in contract wording and system behavior, periodic review alone is unlikely to catch them.

The third gap is limited transparency. Customers of Islamic fintech can rarely verify whether the contract they signed was ever reviewed and approved by a Shariah authority. Regulators such as the Financial Services Authority (OJK) set operational requirements for technology-based financing providers (OJK, 2022), but they rely largely on reports written by the providers themselves. Without a shared record that anyone can verify, trust among customers, scholars, platforms, and supervisors stays thin, which in turn holds back the financial inclusion that Islamic fintech is meant to promote. Regulatory technology (RegTech) and supervisory technology (SupTech) can help close this gap by automating compliance monitoring (Arner et al., 2016).

Table 1. Compliance gaps and the SHARIAGUARD response

Gap	Consequence	SHARIAGUARD response
Manual, inconsistent checking	Slow review; uneven results across reviewers	Layer 1: AI and NLP screening against fatwa
Hidden non-compliance	Undetected <i>riba</i> , <i>gharar</i> , or altered terms	Layer 2: smart contract locks approved terms
Limited transparency	Weak trust and difficult supervision	Layer 3: audit dashboard with immutable log

3. Conceptual Framework

The framework rests on four ideas. The first is *Maqasid al-Shariah*, the higher objectives of Islamic law, which include the protection of wealth and the realization of justice

and public benefit. Scholars of Islamic economics argue that financial practice should be assessed not only by the form of contracts but also by whether it serves these objectives (Chapra, 2008; Dusuki & Abdullah, 2007). For SHARIAGUARD, this means that automation must serve fairness and transparency and not merely speed.

The second idea is the *akad* itself. A valid contract requires identifiable parties, a lawful subject matter, and a clear offer and acceptance, together with the specific conditions of each contract type. Because these conditions can be expressed as rules and clause patterns, they provide a workable basis for partial automation.

The third idea is digital Shariah governance, which means embedding Shariah supervision into the technology lifecycle of a platform rather than treating it as a separate, periodic review. Fintech innovation reshapes how financial services are delivered (Gomber et al., 2018), and governance must therefore move at a similar pace.

The fourth idea is ethical AI. Floridi et al. (2018) propose five principles for a good AI society: beneficence, non-maleficence, autonomy, justice, and explicability. This essay translates them into three operational requirements for Islamic fintech: fairness, so that the system does not disadvantage particular customers or interpretations; transparency, so that each output can be explained; and accountability, so that a responsible human decision-maker is always identifiable.

On the technical side, SHARIAGUARD draws on pre-trained language models for text understanding (Devlin et al., 2019), retrieval-augmented generation for grounding outputs in a verified knowledge base (Lewis et al., 2020), and permissioned blockchain platforms for controlled and auditable record keeping (Androulaki et al., 2018).

4. Design of the SHARIAGUARD Framework

SHARIAGUARD is designed as a three-layer framework that moves compliance control from periodic manual review to continuous, automated, and auditable supervision. Each layer addresses one gap identified in Section 2, and a Shariah board review stage keeps human scholars as the final authority. Figure 1 shows the overall flow.

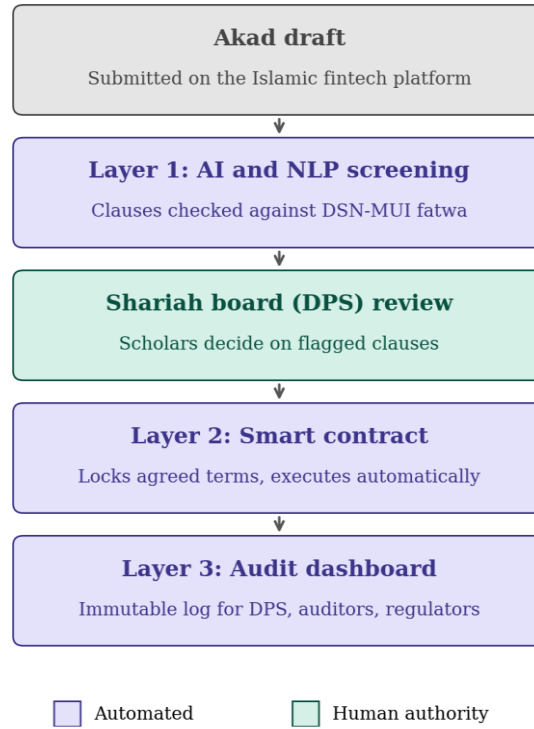


Figure 1. SHARIAGUARD system flow

4.1 Layer 1: AI and NLP screening module

The first layer examines every *akad* draft before it is signed. A natural language processing model extracts key clauses from the contract, such as the profit-sharing ratio in *mudharabah*, margin disclosure in *murabahah*, and provisions for late payment. These clauses are then compared with a structured knowledge base built from DSN-MUI fatwas. A retrieval-augmented approach allows the system to cite the specific fatwa behind each finding, so that the output is a traceable reference and not an opaque verdict. Clauses that resemble prohibited elements, such as *riba*, *gharar*, or *maysir*, are flagged together with a risk level and a short explanation.

4.2 Shariah board review

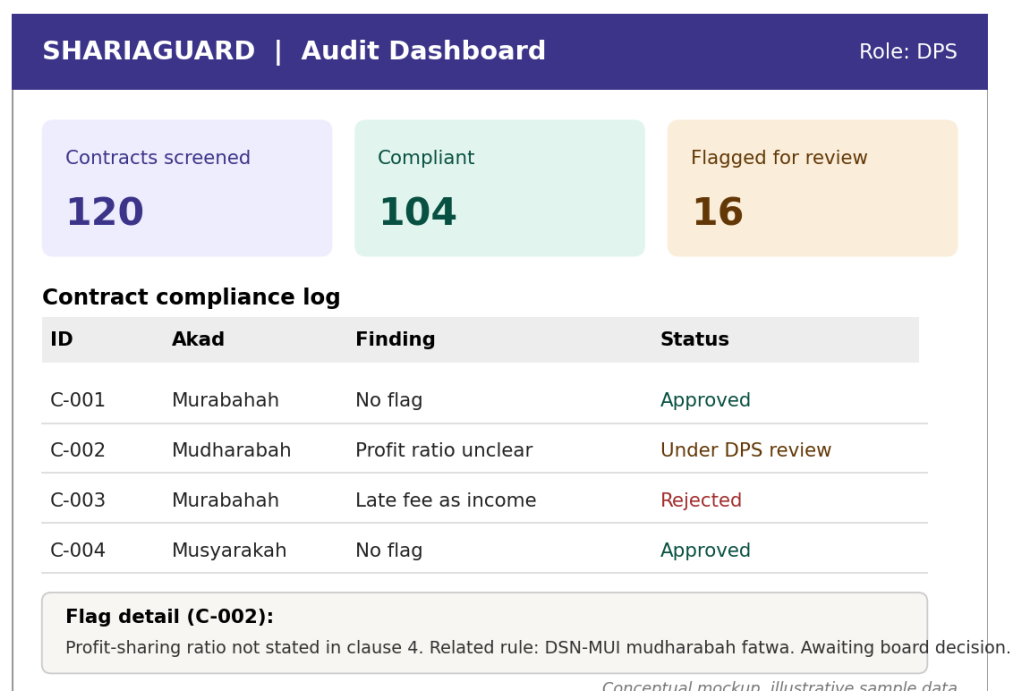
Flagged clauses are routed to the DPS through a review queue. For each item, the board can approve, amend, or reject the clause, and every decision is stored as a precedent that makes later screening more accurate. A contract with no flagged clauses moves straight to the next stage, which lets scholars spend their time on the difficult cases instead of routine ones. This reflects the human-in-the-loop principle: the AI assists with screening, but only the board can issue a Shariah ruling.

4.3 Layer 2: Smart contract enforcement

Once the board has approved an *akad*, its essential terms are encoded in a blockchain-based smart contract. The agreed profit-sharing ratio, the fixed margin, or a ban on extra charges are locked at signing, and payments follow those terms. Because no single party can change the contract afterwards, the platform has much less room to alter terms after approval. A permissioned blockchain keeps sensitive customer data protected while still allowing authorized parties to validate records together.

4.4 Layer 3: Audit dashboard

The third layer is a real-time dashboard for the DPS, internal auditors, and regulators. It shows the compliance status of every contract, the history of flagged clauses, the board's decisions, and the transaction records held on the ledger. Because those records cannot be edited retroactively, supervisors get a dependable audit trail and can complete their checks faster. Customers also see a simplified compliance status for their own contracts, which gives them a concrete reason to trust the platform.



SHARIAGUARD is not only a piece of software. It is a package that combines a governance model, tools, a policy proposal, a program, and a new habit, as summarized in Table 2. This combination matters because technology alone does not change supervisory practice unless rules, institutions, and user behavior change with it.

Table 2. The SHARIAGUARD solution package

Form	Component	Description
Model	3-layer governance	AI screening, scholar review, and on-chain enforcement
Tool	Screening engine and dashboard	Fatwa-grounded NLP and real-time audit view
Policy	Machine-readable fatwa standard	Structured digital format for DSN-MUI fatwas
Program	SANDI-SYARIAH	Regulatory sandbox for AI in Islamic finance
Habit	Check before you sign	Customers view compliance status before accepting an akad

5. Ethical AI and Shariah Governance

What distinguishes SHARIAGUARD from a generic application of AI in finance is the set of safeguards built into its design.

The first safeguard is scholarly authority. The system never issues a final ruling: it prepares the evidence and the DPS decides. If scholars override an AI finding, the override prevails and is logged together with the reason. This keeps the power to interpret Shariah in the hands of qualified scholars and leaves accountability with identifiable people.

The second safeguard is explainability. Every flag must show the clause in question, the fatwa or rule it relates to, and a plain-language reason. Research on explainable AI stresses that users need to understand and challenge model outputs before a system can be used responsibly (Arrieta et al., 2020). Here explainability is also a matter of religious trust, since a scholar cannot responsibly rely on a verdict that cannot be examined.

Third, the framework treats fairness as respect for differing interpretations. Fatwas and scholarly opinions often differ across institutions and jurisdictions, so hard-coding a single view would favor it and disadvantage institutions that follow another. SHARIAGUARD therefore lets each institution configure its own rule set under its DPS, and it reports uncertainty or disagreement instead of forcing a yes-or-no answer. The training and evaluation data should also be checked for bias, for instance where contracts from small or rural providers are underrepresented.

Fourth comes data protection. Personal data should never be written to a public or shared ledger, so only cryptographic hashes and compliance status are recorded on-chain.

Personal and financial details stay in secured off-chain storage that complies with Indonesia's Personal Data Protection Law (Republic of Indonesia, 2022). Dashboard access is role-based, so each party sees only what its function requires.

Finally, the framework has limits that should be stated openly. It cannot judge the intention behind a transaction, nor can it detect structures that follow the form of a valid contract while violating its substance. These matters require scholarly assessment, which is one reason the board review stage remains essential. Acknowledging such limits is part of using AI ethically.

6. Implementation and Feasibility

SHARIAGUARD can be implemented in three phases. In the first, a student-built prototype uses a fine-tuned multilingual language model for clause extraction, a vector database of digitized fatwas for retrieval, and a smart contract on a permissioned test network such as Hyperledger Fabric. The prototype is then tested on sample contracts that Shariah experts have labeled in advance. In the second phase, one Islamic fintech platform runs a pilot under the supervision of its DPS, ideally inside a regulatory sandbox that we propose to call SANDI-SYARIAH. The pilot is judged on three measures: how closely the AI flags match expert judgement, how much review time is saved, and what share of contracts end up with a complete audit trail. In the third phase, the refined framework is offered to other platforms and supervisors.

Several obstacles are likely. The legal status of smart contracts and electronic records still has to be clarified with regulators, so early talks with OJK and DSN-MUI are necessary. High-quality labeled data is scarce, and working with scholars and universities to build a shared, curated collection of clauses and fatwa references could fill that gap. Scholars and platform staff may also have limited digital skills, which calls for training and a simple interface. Costs can be kept manageable by starting with a few common contract types and using open-source tools, which also keeps the framework within reach of smaller providers that serve MSMEs.

SHARIAGUARD has educational value too. The structured fatwa knowledge base and the annotated clause examples could be used to teach students of Islamic economics and law, and to train new supervisory staff. In that sense, the framework helps improve how Shariah finance is taught, alongside how compliance is practiced.

7. Contribution to the Sustainable Development Goals

SHARIAGUARD contributes to three goals, summarized in Table 3. The strongest link is to SDG 16, which calls for effective, accountable, and transparent institutions: a verifiable record of Shariah compliance makes financial institutions answerable to both customers and regulators. The framework relates to SDG 9 through its use of new technology in financial infrastructure, and to SDG 8 by helping widen safe, trusted access to financial services, which matters most for micro and small enterprises.

Table 3. Alignment of SHARIAGUARD with the SDGs

SDG	Target	SHARIAGUARD contribution
16 (primary)	16.6 Effective, accountable, transparent institutions	Auditable record of Shariah compliance
9 (supporting)	9.5 Technological upgrading and innovation	AI and blockchain in financial infrastructure
8 (supporting)	8.10 Access to financial services	Trusted digital Islamic financing for MSMEs

8. Conclusion

Islamic fintech can only fulfill its promise if customers and regulators can trust that every contract is compliant. This essay has proposed SHARIAGUARD, a framework that combines AI-based screening, smart contract enforcement, and an auditable dashboard, with the Shariah board keeping the final word. By grounding the technology in *Maqasid Shariah* and in fairness, transparency, and accountability, the framework offers a practical path to continuous, verifiable Shariah governance, one that advances SDGs 16, 9, and 8 in the spirit of Society 5.0.

Three recommendations follow. Regulators and DSN-MUI should support turning fatwas into structured, machine-readable form and provide a sandbox for testing. Fintech providers should build Shariah review into their systems from the start. Academics and student researchers should create open datasets and evaluate these systems rigorously, so that innovation in Islamic finance stays both effective and faithful to its ethical foundations.

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